On the Informativeness of Moments in Optimal Stopping

José Correa
Universidad de Chile
Department of Industrial Engineering
Santiago, Chile
correa@uchile.cl

Andrés Cristi
EPFL
College of Management of Technology
Lausanne, Switzerland
andres.cristi@epfl.ch

Vasilis Livanos
Universidad de Chile
Center for Mathematical Modeling
Santiago, Chile
vas.livanos@gmail.com

Victor Verdugo
PUC Chile
Institute for Mathematical and Computational Engineering and School of Engineering
Santiago, Chile
victor.verdugo@uc.cl

Jiechen Zhang
EPFL
College of Management of Technology
Lausanne, Switzerland
jiechen.zhang@epfl.ch

Abstract

We study a variant of the prophet inequality with limited information, where the decision maker has access only to the first k moments of each random variable, rather than their full distributions. In this work, we show that even with full moment knowledge (i.e., $k = \infty$), the best possible competitive ratio is $\Theta(1/\log n)$, and that this can already be achieved with only knowledge of the first moment. While the lower bound is simple and is attained by a standard exponential bucketing algorithm, the upper bound requires a subtle construction. This involves using Vandermonde matrices first to construct a parametrized family of distributions for which the first k moments coincide, and for which the expected maximum of n such copies varies widely across different parameter choices. Using Prokhorov's theorem, we establish the existence of limit distributions, which we show have all their moments equal. Finally, we describe a construction where an adversary can select *equally looking* instances combining these distributions, making it impossible for the decision maker to obtain a factor better than $O(1/\log n)$ of the expected maximum.

Our result implies that to obtain improved prophet inequalities, further assumptions beyond moment knowledge are needed. To showcase this direction, we establish improved bounds under additional distributional assumptions such as MHR and bounded coefficient of variation.

CCS Concepts

• **Theory of computation** → **Algorithmic mechanism design**; **Online algorithms**.

Keywords

prophet inequality, limited information, optimal stopping, moments



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1 Introduction

The prophet inequality is a cornerstone of online decision-making and optimal stopping theory, and an essential tool in the theory of posted-price mechanisms. In its classical setting, a decision maker is presented with a sequence of n independent random variables, $X_1, X_2, \ldots, X_n$ drawn from known distributions. The value of each X_i is revealed one by one, and upon seeing it, the decision maker must make an immediate and irrevocable decision: either accept the current value and stop, or reject it and continue to the next. The goal is to devise a stopping rule, which we call ALG, that maximizes the expected reward. Krengel, Sucheston and Garling [18] first devised a stopping rule that guarantees an expected outcome of at least half of the expected maximum; that is, $\mathbb{E}[\text{ALG}] \geq \frac{1}{2}\mathbb{E}[\max_i X_i]$, and showed that this is the best guarantee possible in the worst-case. Later, in a celebrated result, Samuel-Cahn [24] showed that a surprisingly simple threshold-based strategy can achieve the same optimal guarantee.

This foundational result, however, relies on a strong assumption: that the decision-maker knows the exact distribution of each random variable. In many real-world scenarios, from resource allocation to online auctions, this level of detail is a luxury we do not have. This raises natural and practical questions related to the attainable guarantees when only partial information about the distributions is disclosed to the decision maker. A rich line of research on prophet inequalities under informational constraints has recently tackled these questions, exploring models where the decision-maker has sample access to the distributions [1, 3, 8, 13, 23] or possesses priors that may be inaccurate [10].

In this work, we take a different perspective on this question by introducing an alternative yet natural model of partial information: One where the decision maker knows only a number of moments for each distribution. Specifically, we assume that for

each random variable X_i, we are given the first k moments, $\mathbb{E}\left[X_i^j\right]$ for $j = 1, \ldots, k$ as input, but not the underlying distribution itself. Thus, the question becomes:

> *What guarantees can we get by knowing only the first k moments of each distribution?*

This moment-based informational model is natural and well-motivated. In many real-world situations, due to privacy, proprietary, or computational reasons, data owners are willing to disclose only coarse statistics – such as means, variances, or higher-order moments – rather than raw samples or whole datasets. For example, an online advertising exchange may reveal to bidders the average click-through rate (CTR) and variance of each ad slot (to reassure them of traffic quality) but refuse to share user-level impression data for privacy and competitive reasons. The decision maker must then act using only these published moments, which makes our problem practically relevant. In addition, model-based decision-making typically relies on summary statistics, rather than complete distributions, and models are generally perceived to be indicative of the underlying data.

Moment-based models have recently gained attention in auction theory, where several works study the design of robust auctions when only a few moments of the valuation distributions are known; see, e.g., [2, 4, 5, 26]. Similarly, in the field of distributionally robust optimization, an important line of work consists of optimizing over decisions that are resilient to the worst-case distribution matching a given set of moments [9]. This perspective remains largely unexplored in online optimal stopping problems. Our work seeks to understand the power and fundamental limitations of this informational model in this domain. In particular, our results settle an open question first formulated in a survey by Correa et al. [7].

1.1 Our Results

To place our main result into context, we first investigate the fundamental case of knowing only the first moment ($k = 1$). In this case, we establish tight bounds, up to constant factors, on the competitive ratio: we design a randomized algorithm (Algorithm 1) that achieves a guarantee of $\Omega(1/\log n)$ (Theorem 3.1) and show that no algorithm can achieve a competitive ratio better than $O(1/\log n)$ (Theorem 3.2).

Given these results, one may wonder whether knowing more than one moment leads to better guarantees on the competitive ratio. This is because standard tail inequalities that are often associated with moment knowledge (e.g., Markov's and Chebyshev's inequalities) lead to a natural belief that more and higher-order moments allow for better control over the tails of the distributions, and thus for better control over the expected maximum. Perhaps counterintuitively, our strong negative result illustrates that this belief is incorrect. Our main contribution is the following:

Main Result. *Even if $k = \infty$, there is no algorithm achieving a factor better than $O(1/\log n)$ (Theorem 5.1). We achieve this by constructing adversarial distributions with matching moments but for which $\mathbb{E}[\max_i X_i]$ has vastly different behaviour.*

We make a couple of remarks regarding our results. First, it is worth noting that in the case of $k = \infty$, our problem does not become equivalent to the standard known-distributions case. In fact, there exist distributions that share *all* moments (and all are finite) but are *not* the same distribution [14]. Given this, the reader may wonder whether our negative results only hold due to quirky distributions that are moment-indeterminate. However, this is not the case as, in the process of constructing the moment-matching distributions for $k = \infty$, we also construct for every finite k and n an instance with bounded-support (and thus moment-determinate) distributions for which the $O(1/\log n)$ impossibility continues to hold.

Our results establish that only moment information about the underlying distributions is not sufficient to break the logarithmic barrier, which, in turn, is easily achievable even with one moment. Thus, we complement our main results by showcasing how additional information can enhance the competitiveness guarantee. First, for the family of absolutely continuous distributions with monotone hazard rate (MHR), using only first-moment knowledge, we show that our exponential bucketing algorithm achieves an improved competitive ratio of $\Omega(1/\log\log n)$ (Section 6.1). Second, when $k = \Omega(\log n)$, under some additional condition on the moments which restricts their speed of growth, we design a single-threshold algorithm that achieves a constant competitive ratio (Section 6.2).

1.2 Technical Overview of Our Main Result

Our algorithmic result relies on a relatively simple and well-known idea: *exponential bucketing*. Specifically, we use the first moments to obtain upper and lower bounds that confine the expected maximum to a (multiplicatively) linear range. Then, we use an exponential bucketing strategy to guess the expected maximum within a constant range, noting that, conditioned on our guess being correct, we recover a constant factor guarantee on the competitive ratio.

On the other hand, the main technical ingredients of our negative results are far more involved. As it turns out, the difficulty of this setting stems from the existence of multiple, distinct distributions that share the same first k moments but yield vastly different values for $\mathbb{E}[\max_i X_i]$. This moment-matching ambiguity makes it difficult to select a single, robust stopping threshold (or even multiple adaptive thresholds), since a value that seems promising under one possible distribution could be highly suboptimal for another.

Our approach to showing our negative result consists of two stages. We begin by considering a finite but arbitrarily large k, and leverage the rapid growth of the moments of long-shot-type random variables with many low probability events to construct different distributions for which the first k moments match exactly, but for which the expected maxima of n i.i.d. copies are a factor n away. To do this, we write a linear system for the existence of these distributions and their matching moments that involves Vandermonde matrices, commonly used in polynomial interpolation, then leverage their full-rank properties to show that the system admits a solution. Already, the existence of such distributions shows that exponential bucketing cannot obtain better-than-logarithmic guarantees.

To obtain our negative result for $k = \infty$, we prove that for any n, all moment-matching distributions that we constructed previously have well-defined limits as $k \to \infty$. These limiting distributions have *all moments* equal, but crucially maintain their property of

having expected maxima of n i.i.d. copies a factor n away. In other words, we show the existence of a *universal moment sequence* which allows the expected maxima of n i.i.d. copies drawn from distributions with this moment sequence to be a factor n away. To complete our negative result, we combine several different distributions of this type into $\Omega(\log n)$ different instances that cannot be effectively distinguished by any algorithm. In this way, we extend the impossibility from the exponential-bucketing algorithm to any algorithm.

1.3 Further Related Work

In prior work, the question of modeling the prophet inequality under limited information has been addressed mainly under the sample-based model, pioneered by the work of Azar et al. [1]. In this model, we are given access to each distribution through independent samples, and there are two central questions: (i) how many samples are enough to achieve similar guarantees to the standard case, and (ii) given few samples per distribution (e.g., one or two), what guarantees can be obtained? Notably, in the standard fixed-order model, a single sample per distribution is enough to guarantee the tight $1/2$-approximation, which answers both questions at once [23].

On the other hand, for the i.i.d. variant, n samples (one "per distribution") give only a 0.671-approximation, and to approach the tight $0.745 - \varepsilon$ approximation, we require $O(n/\varepsilon)$ many samples ($O(1/\varepsilon)$ per distribution) [6]. Interestingly, it was recently shown that a constant number of samples per distribution is enough to approach the optimal guarantee for *all* single-selection variants of the prophet inequality [8]. These questions have also been considered in a wide range of combinatorial variants of the prophet inequalities; see, e.g., [3, 11, 12, 20].

Another important but less studied model is the one with inaccurate priors [10]. There, the algorithm is given distributions, but they are not necessarily the true distributions. Instead, the true distributions are guaranteed to be close to the given distributions, under a given metric. Notably, for most standard metrics in the space of distributions, an ε distance guarantees a loss of at most $O(n\varepsilon)$ in the expected reward, assuming all variables lie in $[0, 1]$.

The question of how well can the decision maker do in the prophet inequality with only moment knowledge was first formulated in the survey by Correa et al. [7]. Recently, Kleer and van Leeuwaarden [15] studied threshold algorithms for the prophet inequality using only the first two moments. They did not analyze the problem in terms of the competitive ratio, but instead focused on understanding the structure of the optimal robust algorithm.

In the literature on robust mechanism design, considering ambiguity sets defined by the first moment or the first few moments is also a recent approach. Carrasco et al. [4] studied the problem of a single seller that sells an item to a single buyer with a valuation drawn from a distribution. The seller maximizes the revenue, but only knows the first k moments of the distribution. Other works have considered the case of 2 or more buyers under similar assumptions [2, 5]. These works focus on understanding the structure of the optimal robust mechanism. Closer to us, Wang et al. [26] studied the approximation ratio for the case of a single seller and a single buyer, under knowledge of the first two moments. They showed that the approximation ratio is unbounded, but decays as $\Theta(1/\log(\sigma/\mu))$.

2 Model

In the *k-moment prophet inequality* problem, an adversary decides on n distributions $F_1, \ldots, F_n$ supported in $\mathbb{R}_{\geq 0}$. The decision maker does not have full knowledge of the distributions F_i but has access to the first k moments of each distribution. In other words, for every $i \in [n], j \in [k]$, the decision maker is given $\mathbb{E}\left[X_i^j\right]$. We assume the adversary is restricted to distributions for which the first k moments are finite.

The decision maker observes sequentially the realizations of n independent random variables $X_i \sim F_i$, for $i \in [n]$. Immediately upon seeing X_i the decision maker must irrevocably decide to either stop and receive reward X_i, or continue to the next stage. The objective of the decision maker is to maximize the selected value and the benchmark they are compared to is the value obtained by a prophet with a priori access to all realizations. If we denote the expected value of an algorithm ALG by $\mathbb{E}[\text{ALG}]$ and the expected value of the prophet by $\mathbb{E}[\text{OPT}] \coloneqq \mathbb{E}[\max_i X_i]$, we say that a stopping algorithm ALG is α-competitive for some $\alpha \in [0, 1]$ if, for every set instance of the problem, we have $\mathbb{E}[\text{ALG}] \geq \alpha \cdot \mathbb{E}[\text{OPT}]$. The ratio $\mathbb{E}[\text{ALG}] / \mathbb{E}[\text{OPT}]$ is called the competitive ratio.

We sometimes use $\mu^\star = \max_i \mathbb{E}[X_i]$ to denote the maximum expectation, $(x)^+$ to denote $\max\{x, 0\}$ and $\mathbb{I}(\mathcal{E})$ for the indicator of the event $\mathcal{E}$ which is equal to 1 if $\mathcal{E}$ happens and 0 otherwise.

3 The Known Expectation Case ($k = 1$)

Perhaps the most natural first question is what can be achieved when the decision maker is only given access to the mean μ_i of each F_i, i.e., the $k = 1$ case. We start with the following observation.

Observation 3.1. *Let $X_1, \ldots, X_n$ be non-negative random variables with $\mu^\star > 0$ and $n \geq 2$. Then, we have $\mu^\star \leq \mathbb{E}[\text{OPT}] \leq n\mu^\star$.*

Proof. The lower bound comes from noting that for all $j \in [n]$, we have $X_j \leq \max_i X_i$. The upper bound is a consequence of the inequality $\max_i X_i \leq \sum_{i=1}^n X_i$ when all values are non-negative, and of the linearity of the expectation. □

Next, we present an algorithm that achieves a logarithmic competitive ratio using only the first moment of each distribution. The main idea behind our algorithm is an exponential bucketing trick that allows us to "guess" the correct scale of $\mathbb{E}[\text{OPT}]$, which happens with logarithmic probability and, conditional on the guess being correct, achieves a constant approximation to $\mathbb{E}[\text{OPT}]$.

Algorithm 1

Draw $j \sim \text{Unif}\{0, \ldots, \lceil \log_2 n \rceil - 1\}$ and let $\tau \leftarrow 2^{j-1}\mu^\star$
for $i = 1, 2, \ldots, n$ **do**
  if $X_i \geq \tau$ **then**
    accept X_i and **stop**

Theorem 3.1. *Algorithm 1 accepts a value* ALG *such that*

$$\mathbb{E}[\text{ALG}] \geq \mathbb{E}[\text{OPT}] / (4\lceil \log_2 n \rceil).$$

Proof. Let $j^\star \in \{0, 1, \ldots, \lceil \log_2 n \rceil - 1\}$ be such that $2^{j^\star}\mu^\star \le \mathbb{E}[\mathrm{OPT}] \le 2^{j^\star+1}\mu^\star$. By Observation 3.1, such $j^\star$ exists. Also, let $\tau^\star = 2^{j^\star-1}\mu^\star$. This implies that

$$2\tau^\star \le \mathbb{E}[\mathrm{OPT}] \le 4\tau^\star. \tag{1}$$

We have $\mathbb{E}[\mathrm{ALG}] \ge \Pr\left[j = j^\star\right] \cdot \mathbb{E}\left[\mathrm{ALG} \mid j = j^\star\right] = \mathbb{E}\left[\mathrm{ALG} \mid j = j^\star\right]/\lceil \log_2 n \rceil$, since $j = j^\star$ and thus $\tau = \tau^\star$ happens with probability $1/\lceil \log_2 n \rceil$. Next, we show that when $j = j^\star$, meaning the algorithm sets $\tau = \tau^\star$, it gets a constant approximation to $\mathbb{E}[\mathrm{OPT}]$. The proof follows from [16, 27], but we include it for completeness. Let $p = \Pr\left[\max_i X_i \ge \tau^\star\right]$. We have

$$\begin{aligned}
\mathbb{E}\left[\mathrm{ALG} \mid j = j^\star\right] &= p\tau^\star + \sum_{i=1}^{n} \Pr[X_j < \tau^\star \text{ for all } j < i] \\
&\qquad \cdot \mathbb{E}\left[(X_i - \tau^\star)^+\right] \\
&\ge p\tau^\star + (1-p)\sum_{i=1}^{n} \mathbb{E}\left[(X_i - \tau^\star)^+\right] \\
&\ge p\tau^\star + (1-p)\mathbb{E}\left[\max_{i=1,\ldots,n}(X_i - \tau^\star)^+\right] \\
&\ge p\tau^\star + (1-p)\left(\mathbb{E}[\mathrm{OPT}] - \tau^\star\right) \\
&\ge \min\left\{\tau^\star, \mathbb{E}[\mathrm{OPT}] - \tau^\star\right\}.
\end{aligned}$$

By (1), it follows that $\min\left\{\tau^\star, \mathbb{E}[\mathrm{OPT}] - \tau^\star\right\} \ge \mathbb{E}[\mathrm{OPT}]/4$, and thus $\mathbb{E}\left[\mathrm{ALG} \mid j = j^\star\right] \ge \mathbb{E}[\mathrm{OPT}]/4$, which concludes the proof. □

We remark that the constant factor in $O(1/\log n)$ can be improved to an optimal one via a continuous analogue of bucketing in which the threshold is chosen from a log-uniform distribution over scales.

Next, we show that it is not possible to beat the competitive ratio of $O(1/\log n)$ if the algorithm has access to only the first moment of each random variable.

Theorem 3.2. *For every $n \ge 2$, there exists a family of instances of the 1-moment prophet inequality problem with n random variables such that for every (deterministic or randomized) stopping rule* ALG *we have*

$$\mathbb{E}[\mathrm{ALG}] \le O\left(\frac{1}{\log n}\right)\mathbb{E}[\mathrm{OPT}].$$

Proof. Let $\ell \in \mathbb{N}$ such that $n = 2^{\ell+1} - 1$, appropriately padding the instance with deterministic zero-valued random variables at the end if there is no such ℓ, and notice that $\ell = \Theta(\log n)$. For every $i \in \{0, 1, \ldots, \ell\}$ define the Bernoulli random variable B_i with probability $1/2^i$, and zero otherwise; in particular $B_0 = 1$ with probability 1. Notice that $\mathbb{E}[B_i] = 1$ for all i. Instantiate the following $\ell + 1$ sequences, indexed by $i = 0, 1, \ldots, \ell$, all of length n, with blocks of i.i.d. copies of the Bernoulli variables:

$$\mathrm{Seq}_i: \quad \underbrace{B_0}_{1 \text{ copy}}, \ \underbrace{B_1, B_1}_{2 \text{ copies}}, \ \ldots, \ \underbrace{B_i, \ldots, B_i}_{2^i \text{ copies}}, \ \ldots, \ \underbrace{B_0, \ldots, B_0}_{n - \sum_{j=1}^{i} 2^j \text{ copies}}.$$

For every i, since $\max_j X_j \le 2^i$, we have

$$(1 - 1/e)2^i \le \mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right] \le 2^i, \tag{2}$$

where the probability that there is a non-zero X_j among the 2^i copies of B_i is $1 - \left(1 - 1/2^i\right)^{2^i} \ge 1 - 1/e$.

Fix an algorithm ALG, and notice that, since all variables have mean 1, the $\ell + 1$ instances are indistinguishable at the beginning. Let α_j be the probability (over its internal randomness) that, on instance Seq_ℓ, ALG accepts a value in the j-th block, i.e., from one of the 2^j copies of B_j. These correspond to disjoint events, so $\sum_{j=1}^{\ell} \alpha_j \le 1$. Furthermore, since the blocks arrive in increasing order of magnitude in all instances $\mathrm{Seq}_0, \ldots, \mathrm{Seq}_\ell$, upon observing values from block j, the algorithm does not know which of the instances from $\mathrm{Seq}_j, \ldots, \mathrm{Seq}_\ell$ it is observing. Therefore, for all instances Seq_i where $i \ge j$, the probability of accepting a value from the j-th block is also α_j. Thus, we have

$$\begin{aligned}
\mathbb{E}\left[\mathrm{ALG}(\mathrm{Seq}_i)\right] &= \sum_{j=0}^{i} \alpha_j \mathbb{E}\left[\mathrm{ALG}(\mathrm{Seq}_i) \mid \text{first accept is in } B_j\right] \\
&\le \sum_{j=0}^{i} \alpha_j 2^j = 2^i \cdot \sum_{j=0}^{i} \alpha_j 2^{j-i} \\
&\overset{(2)}{\le} \frac{\sum_{j=0}^{i} \alpha_j 2^{j-i}}{1 - 1/e} \cdot \mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right].
\end{aligned}$$

The adversary picks the instance with the minimum competitive ratio, which is lower than the average competitive ratio over the $\ell + 1$ instances. Thus, the competitive ratio is at most

$$\begin{aligned}
\frac{1}{\ell+1}\sum_{i=0}^{\ell} \frac{\mathbb{E}\left[\mathrm{ALG}(\mathrm{Seq}_i)\right]}{\mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right]} &\le \frac{1}{\ell+1}\sum_{i=0}^{\ell} \frac{\sum_{j=0}^{i} \alpha_j 2^{j-i}}{1 - \frac{1}{e}} \\
&= \frac{1}{1 - \frac{1}{e}} \cdot \frac{1}{\ell+1}\left(\sum_{j=0}^{\ell} \alpha_j \sum_{r=0}^{\ell-j} 2^{-r}\right) \\
&\le \frac{1}{1 - 1/e} \cdot \frac{1}{\ell+1}\sum_{j=0}^{\ell} \alpha_j \left(\sum_{r=0}^{\infty} 2^{-r}\right) \\
&\le \frac{2}{1 - 1/e} \cdot \frac{1}{\ell+1} = O\left(\frac{1}{\ell}\right),
\end{aligned}$$

where the last inequality follows from $\sum_{j=0}^{\ell} \alpha_j \le 1$. Since $\ell = \Theta(\log n)$, the theorem follows. □

4 Finite Moment Knowledge ($1 < k < \infty$)

In this section, we generalize the $O(1/\log n)$ upper bound on the competitive ratio we showed above to instances where k is any natural number, even if it grows with n. The argument is structured as follows. We begin by describing a useful tool in Proposition 4.1, which shows how to modify a discrete distribution's support while preserving its first k moments. We then use this lemma to prove Lemma 4.1, which formally constructs a family of distributions $F^{(k)}_{\alpha,n}$. These distributions are the cornerstone of our impossibility result: they are indistinguishable based on their first k moments, yet as we show in Proposition 4.2, the expected maximum of n i.i.d. samples from them can be tuned by a parameter $\alpha \in (1, 2)$ to vary over a wide range. This behaviour demonstrates why exponential bucketing strategies fail. Finally, with these distributions and a key estimate from Lemma 4.2, we present the proof of our main result, Theorem 4.1, which uses a sophisticated adversarial construction to generalize this failure to *any* online algorithm.

Theorem 4.1. *For every $n \ge 2$ and any $k > 1$, there exists a family of instances of the k-moment prophet inequality problem with*

n random variables such that for every (deterministic or randomized) stopping rule ALG*, we have*

$$\mathbb{E}[\mathrm{ALG}] \le \mathrm{O}\left(\frac{1}{\log n}\right)\mathbb{E}[\mathrm{OPT}].$$

To build up to the proof of Theorem 4.1, we start by constructing families of i.i.d. instances that agree on the first k moments, yet their expected maxima are essentially anywhere in the range (n, n^2). This construction demonstrates that even when we know any number of moments, the expected maximum is still essentially unconstrained, implying that there is no hope that the exponential bucketing strategy can achieve a competitive ratio better than $\Theta(1/\log n)$. Then, we show how to use these distributions to rule out improvements for *any* online algorithm.

To construct the families of distributions, we first construct auxiliary distributions $\tilde{F}_n^{(k)}$ as follows. For $\ell \ge 2$, let $w_\ell = n^{2(\ell-1)}$ and $q_\ell = 1/(n^{\ell^2-1})$. For $X \sim \tilde{F}_n^{(k)}$, we define

$$X = \begin{cases} w_\ell & \text{w.p. } q_\ell, \quad \text{for } \ell = 2,3,\dots,k \\ 1 & \text{w.p. } \frac{1}{2} - \sum_{\ell=2}^{k} q_\ell \\ 0 & \text{w.p. } \frac{1}{2} \end{cases} \tag{3}$$

We define $F_{\alpha,n}^{(k)}$ as the resulting distribution when we replace w_2 with n^α and we adjust the probabilities so that the first k moments remain the same. The following lemma establishes that $F_{\alpha,n}^{(k)}$ is a well-defined distribution.

Lemma 4.1. *For each $\alpha \in (1,2)$, there is a unique distribution $F_{\alpha,n}^{(k)}$ that has the same first k moments as $\tilde{F}_n^{(k)}$, and is supported in the set $\{0, 1, n^\alpha, w_3, w_4, \dots, w_k\}$. Moreover, if $p_\ell^{\alpha,n,k}$ denotes the probability of w_ℓ under distribution $F_{\alpha,n}^{(k)}$, and $p_2^{\alpha,n,k}$ is the probability of n^α, we have that*

$$p_2^{\alpha,n,k} = \frac{1}{n^{2\alpha-1}}(1+\mathrm{o}(1)), \quad \text{and} \quad p_\ell^{\alpha,n,k} = q_\ell(1+\mathrm{o}(1))$$

for $\ell \in \{0,1,3,\dots,k\}$, where $\mathrm{o}(1)$ is independent of k and ℓ.

To prove Lemma 4.1, we use a technical tool (Proposition 4.1) that allows us to move from one distribution to another, with different support, while preserving the first k moments. Consider the two vectors $\mathbf{x} = [x_0, x_1, \dots, x_k]$ and $\mathbf{y} = [y_0, y_1, \dots, y_k]$ in $\mathbb{R}^{k+1}$. Given a probability distribution $\mathbf{q} = [q_0, q_1, \dots, q_k] \ge 0$ supported on $\mathbf{x}$, our goal is to construct a new probability distribution $\mathbf{p} = [p_0, p_1, \dots, p_k] \ge 0$ supported on $\mathbf{y}$ that has exactly the same first k moments as $\mathbf{q}$, i.e., we have $\sum_{i=0}^k y_i^\ell p_i = \sum_{i=0}^k x_i^\ell q_i$ for every $\ell \in \{0,1,\dots,k\}$. This condition can be written compactly as the following linear system:

$$V_{\mathbf{y}}\mathbf{p} = V_{\mathbf{x}}\mathbf{q}, \tag{4}$$

where $V_{\mathbf{y}}, V_{\mathbf{x}}$ denote the Vandermonde matrices associated with $\mathbf{y}$ and $\mathbf{x}$, namely,

$$V_{\mathbf{y}} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ y_0 & y_1 & \cdots & y_k \\ y_0^2 & y_1^2 & \cdots & y_k^2 \\ \vdots & \vdots & \ddots & \vdots \\ y_0^k & y_1^k & \cdots & y_k^k \end{bmatrix} \text{ and } V_{\mathbf{x}} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_0 & x_1 & \cdots & x_k \\ x_0^2 & x_1^2 & \cdots & x_k^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_0^k & x_1^k & \cdots & x_k^k \end{bmatrix}.$$

The Vandermonde matrix $V_{\mathbf{y}}$ has full rank if and only if all entries of $\mathbf{y}$ are different. In this case, we have $\mathbf{p} = V_{\mathbf{y}}^{-1}V_{\mathbf{x}}\mathbf{q}$. The inverse of a Vandermonde is well-known (see, e.g., [17], eq.(6)):

$$V_{\mathbf{y}}^{-1} = \begin{bmatrix} L_{00} & L_{01} & \cdots & L_{0,k} \\ L_{10} & L_{11} & \cdots & L_{1,k} \\ \vdots & \vdots & \ddots & \vdots \\ L_{k,0} & L_{k,1} & \cdots & L_{k,k} \end{bmatrix},$$

where the entries $L_{i,j}$ are the coefficients of the following polynomial:

$$L_j(x) = \prod_{\substack{0\le s\le k \\ s\ne j}} \frac{x - y_s}{y_j - y_s} = L_{j,0} + L_{j,1}x + \cdots + L_{j,k}x^k.$$

Observe that the polynomial L_j depends on $\mathbf{y}$. Conveniently, the product of the inverse with another Vandermonde matrix can be expressed using the polynomials themselves. Indeed,

$$V_{\mathbf{y}}^{-1}V_{\mathbf{x}} = \begin{bmatrix} L_0(x_0) & L_0(x_1) & \cdots & L_0(x_k) \\ L_1(x_0) & L_1(x_1) & \cdots & L_1(x_k) \\ \vdots & \vdots & \ddots & \vdots \\ L_k(x_0) & L_k(x_1) & \cdots & L_k(x_k) \end{bmatrix}.$$

We are thus led to the following proposition.

Proposition 4.1. *For every distribution $\mathbf{q}$ supported on $\mathbf{x} = [x_0, x_1, \dots, x_k]$, where $x_i \ne x_j$ for all $i \ne j$, and every $\mathbf{y} = [y_0, y_1, \dots, y_k]$ with $y_i \ne y_j$ for all $i \ne j$, there exists a distribution $\mathbf{p}$ supported on $\mathbf{y}$ such that the first k moments of $\mathbf{q}$ and $\mathbf{p}$ are the same. In this case, for every $i \in \{0,1,\dots,k\}$, we have*

$$p_i = \sum_{j=0}^k L_i(x_j)q_j = \sum_{j=0}^k \left(\prod_{\substack{0\le s\le k \\ s\ne i}} \frac{x_j - y_s}{y_i - y_s}\right) q_j. \tag{5}$$

We are now ready to delve into the proof of Lemma 4.1.

Proof of Lemma 4.1. Fix integers $k \ge 2$ and $n \ge 2$, and let $\alpha \in (1,2)$. Define the two vectors given by $\mathbf{x} = [0, 1, w_2, w_3, \dots, w_k]$, and $\mathbf{y} = [0, 1, n^\alpha, w_3, \dots, w_k]$, where $w_\ell := n^{2(\ell-1)}$ for $\ell = 2, \dots, k$. Let the distribution $\mathbf{q}$ be

$$q_0 = \frac{1}{2}, \quad q_1 = \frac{1}{2} - \sum_{\ell=2}^k n^{-(\ell^2-1)}, \quad \text{and} \quad q_\ell = n^{-(\ell^2-1)},$$

for all $\ell = 2, \dots, k$. Let $\mathbf{p} = [p_0^{\alpha,n,k}, p_1^{\alpha,n,k}, p_2^{\alpha,n,k}, \dots, p_k^{\alpha,n,k}]$. For simplicity, we drop the superscripts in the rest of the proof, i.e., we write $p_i = p_i^{\alpha,n,k}$. By Proposition 4.1, for all $i \in \{0,1,\dots,k\}$, we have

$$p_i = \sum_{j=0}^k \left(\prod_{\substack{0\le s\le k \\ s\ne i}} \frac{x_j - y_s}{y_i - y_s}\right) q_j.$$

Since $\mathbf{x}$ and $\mathbf{y}$ coincide at every index except $j = 2$, i.e., $x_j = y_j$ for $j \ne 2$, all terms in the sum cancel, except for $j = i$ and $j = 2$. Hence,

$$p_i = q_i + q_2 \prod_{\substack{0\le t\le k \\ t\ne i}} \frac{n^2 - y_t}{y_i - y_t}, \text{ for } i \ne 2, \text{ and } p_2 = q_2 \prod_{\substack{0\le t\le k \\ t\ne 2}} \frac{n^2 - y_t}{y_2 - y_t}. \tag{6}$$

Then, to show that $|(p_i - q_i)/q_i| = o(1)$ for $i \neq 2$, we show $p_i - q_i = q_2 \prod_{0 \le t \le k : t \neq i} (n^2 - y_t)/(y_i - y_t)$ is small with respect to q_i. Indeed, for p_0, we have

$$\begin{aligned}\left|\frac{p_0 - q_0}{q_0}\right| &= \left|\frac{q_2}{q_0} \cdot \frac{n^2-1}{0-1} \cdot \frac{n^2 - n^\alpha}{0 - n^\alpha} \cdot \prod_{t=3}^{k} \frac{n^2 - n^{2(t-1)}}{0 - n^{2(t-1)}}\right| \\ &\le \frac{q_2}{q_0} \cdot n^2 \cdot \left(1 + O\left(1/n^2\right)\right) \cdot n^{2-\alpha} \cdot \left(1 + O\left(n^{\alpha-2}\right)\right) \\ &= O\left(n^{1-\alpha}\right),\end{aligned}$$

where in the second line, we use the fact that all terms in the product are smaller that one (regardless of k), and in the third line, we used that $q_2 = n^{-3}$ and $q_0 = 1/2$. Since $\alpha > 1$, this is $o(1)$. For p_1, we have

$$\left|\frac{p_1 - q_1}{q_1}\right| = \left|\frac{q_2}{q_1} \cdot \frac{n^2 - 0}{1 - 0} \cdot \frac{n^2 - n^\alpha}{1 - n^\alpha} \prod_{t=3}^{k} \frac{n^2 - n^{2(t-1)}}{1 - n^{2(t-1)}}\right| = O\left(n^{1-\alpha}\right)$$

where the analysis is similar to that of p_0, as all terms are of the same order.

To compute p_i where $i \ge 3$, it is helpful to first define

$$R_{t,i} = \frac{n^2 - y_t}{y_i - y_t}, \quad \text{for } i \neq t.$$

Recall that

$$y_t = \begin{cases} 0, & t = 0, \\ 1, & t = 1, \\ n^\alpha, & t = 2, \\ n^{2(t-1)}, & t \ge 3, \end{cases}$$

and thus, for $i, t \ge 3$,

$$R_{t,i} = \frac{n^2 - n^{2(t-1)}}{n^{2(i-1)} - n^{2(t-1)}}.$$

We analyze $R_{t,i}$ separately for the cases $t < i$ and $t > i$. For $t < i$, we have that

$$\begin{aligned}\left|R_{t,i}\right| &= \frac{n^{2(t-1)} - n^2}{n^{2(i-1)} - n^{2(t-1)}} \\ &= n^{-2(i-t)} \cdot \frac{n^{2(i-1)} - n^{2+2(i-t)}}{n^{2(i-1)} - n^{2(t-1)}} \\ &= n^{-2(i-t)} \cdot \left(1 + \frac{n^{2(t-1)} - n^{2+2(i-t)}}{n^{2(i-1)} - n^{2(t-1)}}\right) \\ &\le n^{-2(i-t)} \cdot \left(1 + 2n^{-2(i-t)}\right),\end{aligned}$$

assuming $n \ge 2$. On the other hand, for $t > i$, we have that

$$\begin{aligned}\left|R_{t,i}\right| = \frac{n^{2(t-1)} - n^2}{n^{2(t-1)} - n^{2(i-1)}} &= 1 + \frac{n^{2(i-1)} - n^2}{n^{2(t-1)} - n^{2(i-1)}} \\ &\le 1 + 2n^{-2(t-i)}\end{aligned}$$

assuming $n \ge 2$. Thus, we have that

$$\begin{aligned}\left|\frac{p_i - q_i}{q_i}\right| &= \left|\frac{q_2}{q_i} \cdot \prod_{t \neq i} R_{t,i}\right| \\ &= \left|\frac{q_2}{q_i} \cdot \frac{n^2 - 0}{n^{2(i-1)}} \cdot \frac{n^2 - 1}{n^{2(i-1)} - 1} \cdot \frac{n^2 - n^\alpha}{n^{2(i-1)} - n^\alpha} \prod_{t \ge 3 : t \neq i} R_{t,i}\right| \\ &= n^{-3} \cdot n^{i^2 - 1} \cdot n^{2-2(i-1)} \cdot n^{2-2(i-1)} \cdot n^{2-2(i-1)} \\ &\quad \cdot (1 + o(1)) \prod_{t=3}^{i-1} |R_{t,i}| \cdot \prod_{t=i+1}^{k} |R_{t,i}| \\ &\le n^{i^2 - 4 + 6 - 6(i-1) + \sum_{t=3}^{i-1} -2(i-t)} \\ &\quad \cdot (1 + o(1)) \prod_{t \neq i} \left(1 + 2n^{-2|i-t|}\right) \\ &\le n^{i^2 - 4 + 6 - 6(i-1) + \sum_{t=3}^{i-1} -2(i-t)} \cdot (1 + o(1)),\end{aligned}$$

where we used our previous computations for $R_{t,i}$, and the fact that $\prod_{j \ge 1}(1 + 2^{1-2j})$ is a constant. Simplifying the exponent of n, we get

$$\begin{aligned}\left|\frac{p_i - q_i}{q_i}\right| &= (1 + o(1)) \cdot n^{i^2 - 6i + 8 - 2\sum_{t=3}^{i-1}(i-t)} = (1 + o(1)) \cdot n^{2-i} \\ &= o(1).\end{aligned}$$

Finally, we compute p_2. Recall that $q_2 = n^{-3}$, and thus

$$\begin{aligned}p_2 &= q_2 \cdot \frac{n^2 - 0}{n^\alpha - 0} \cdot \frac{n^2 - 1}{n^\alpha - 1} \cdot \prod_{t=3}^{k} \frac{n^2 - n^{2(t-1)}}{n^\alpha - n^{2(t-1)}} \\ &= n^{1-2\alpha} \cdot (1 + o(1)) \cdot \prod_{t=3}^{k} \frac{n^{2(t-1)} - n^2}{n^{2(t-1)} - n^\alpha}.\end{aligned}$$

We can upper and lower bound the product in the expression above by $(1 + o(1))$. In fact, for all k,

$$\prod_{t=3}^{k} \frac{n^{2(t-1)} - n^2}{n^{2(t-1)} - n^\alpha} \le 1,$$

and

$$\begin{aligned}\prod_{t=3}^{k} \frac{n^{2(t-1)} - n^2}{n^{2(t-1)} - n^\alpha} \ge \prod_{t=3}^{\infty} \frac{n^{2(t-1)} - n^2}{n^{2(t-1)} - n^\alpha} &\ge \prod_{t=3}^{\infty} \left(1 - \frac{1}{n^{2(t-2)}}\right) \\ &\ge 1 - \sum_{t=3}^{\infty} \frac{1}{n^{2t-4}} \ge 1 - O\left(\frac{1}{n^2}\right).\end{aligned}$$

Therefore, $p_2 = n^{1-2\alpha} \cdot (1 + o(1))$. □

Lemma 4.1 shows how to create distributions $F^{(k)}_{\alpha,n}$ with the same first k moments as $\tilde{F}^{(k)}_n$. Given this, our next goal is to show that the expected maximum of n i.i.d. draws from one of these distributions can be significantly different than that of another. For $\varepsilon > 0$, by taking $\alpha = 1 + \varepsilon/2$ and $\alpha' = 2 - \varepsilon/2$, the following proposition implies that there exist two distributions that share the same first k moments, but their expected maxima of n i.i.d. samples are a factor $\Omega\left(n^{1-\varepsilon}\right)$ apart.

Proposition 4.2. *Suppose that $X_1, \ldots, X_n$ are i.i.d. distributed according to $F^{(k)}_{\alpha,n}$. Then, we have $\mathbb{E}\left[\max_{i=1,\ldots,n} X_i\right] = n^{2-\alpha}(1 + o(1))$, where $o(1)$ is independent of k.*

Proof. To compute the expected maximum, it is essentially enough to consider the contribution of the term n^α. From Lemma 4.1, we have that the probability of this value is $(1 + o(1))/n^{2\alpha-1}$. On the one hand, if we consider the disjoint events $X_i = n^\alpha$, and $X_j \neq n^\alpha$, for all $j \neq i$, we obtain that

$$\mathbb{E}\left[\max_{i=1,\ldots,n} X_i\right] \ge \sum_{i=1}^{n} n^\alpha \cdot \frac{1 + o(1)}{n^{2\alpha-1}} \cdot \left(1 - \frac{1 + o(1)}{n^{2\alpha-1}}\right)^{n-1}$$

$$= n^{2-\alpha}\left(1 - \frac{1+\mathrm{o}(1)}{n^{2\alpha-1}}\right)^{n-1}(1+\mathrm{o}(1))$$
$$\geq n^{2-\alpha}(1+\mathrm{o}(1)).$$

On the other hand, we can split the space into events $\{\max_{i=1,\dots,n} X_i \leq 1\}$ and $\{\max_{i=1,\dots,n} X_i > 1\}$ and use a simple union bound on the latter to obtain

$$\mathbb{E}\left[\max_{i=1,\dots,n} X_i\right] \leq 1 + n\left(n^{\alpha}\cdot\frac{1}{n^{2\alpha-1}} + \sum_{\ell=3}^{k}\frac{n^{2(\ell-1)}}{n^{\ell^2-1}}\right)(1+\mathrm{o}(1))$$
$$\leq 1 + \left(n^{2-\alpha} + \sum_{\ell=3}^{\infty}\frac{n^{2\ell}}{n^{\ell^2}}\right)(1+\mathrm{o}(1))$$
$$\leq n^{2-\alpha}\left(1 + \mathrm{O}\left(\frac{1}{n^{2-\alpha}}\right)\right).$$

Therefore, for $\alpha \in (1,2)$, the expected maximum is $n^{2-\alpha}(1+\mathrm{o}(1))$. □

Proposition 4.2 rules out the possibility of improving the competitive ratio of the exponential bucketing strategy when we have more moments. To extend this impossibility to all algorithms, we employ a construction similar to the one we used for the case $k = 1$ in Theorem 3.2, but we need a slightly stronger version of Proposition 4.2.

Lemma 4.2. *For $n \geq 2$ and $r = n^3$, if we draw r i.i.d. samples $X_1, \dots, X_r$ from the distribution $F^{(k)}_{\alpha,n}$ from Lemma 4.1, then we have*

(1) $\mathbb{E}\left[\max_{i=1,\dots,r} X_i\right] = n^{\alpha}\cdot(1+\mathrm{o}(1))$, *and*
(2) $\mathbb{E}\left[\max_{i=1,\dots,r} X_i \cdot \mathbb{I}\left(X_i > n^{\alpha}\right)\right] \leq \mathrm{O}\left(\frac{1}{n}\right)$,

where the asymptotic terms do not depend on k.

Proof. We start by showing (2) via a union bound.

$$\mathbb{E}\left[\max_{i=1,\dots,r} X_i \cdot \mathbb{I}\left(X_i > n^{\alpha}\right)\right] \leq r\cdot\sum_{\ell=3}^{k} w_\ell q_\ell \cdot (1+\mathrm{o}(1))$$
$$= \sum_{\ell=3}^{k} n^3 \cdot n^{2(\ell-1)} \cdot \frac{1+\mathrm{o}(1)}{n^{\ell^2-1}}$$
$$= \sum_{\ell=3}^{k} n^{2+2\ell-\ell^2}(1+\mathrm{o}(1))$$
$$\leq \frac{1+\mathrm{o}(1)}{n}\sum_{j=0}^{\infty}\frac{1}{2^j} = \mathrm{O}\left(\frac{1}{n}\right),$$

where we used the fact that $2 + 2\ell - \ell^2 \leq 2 - \ell$ for all $\ell \geq 3$, and that $n \geq 2$. Next, we show (1) by computing the probability that $X_i = n^{\alpha}$ for some $i = 1, \dots, r$. Since $\alpha < 2$, we have

$$\Pr\left[X_i = n^{\alpha} \text{ for some } i\right] = 1 - \left(1 - \frac{1+\mathrm{o}(1)}{n^{2\alpha-1}}\right)^{n^3} = 1 - \mathrm{o}(1). \quad \square$$

We are now ready to prove Theorem 4.1.

Proof of Theorem 4.1. The proof follows the structure of the proof of Theorem 3.2. Define $s = \lfloor \frac{1}{4}\log_2 n \rfloor$, and $\ell = \lfloor \frac{3}{4}\log_2 n \rfloor - s$. For $i \in [\ell]$, let α_i be such that $n^{\alpha_i} = n \cdot 2^{s+i}$, that is, $\alpha_i = 1 + (s+i)/\log_2 n$. Let $F_{\alpha_i} := F^{(k)}_{\alpha_i,n}$ be the distribution from Lemma 4.1. Notice that by the definition of s and ℓ, we have that $\alpha_i \in [1.25, 1.75]$, and also there are $\ell = \Omega(\log n)$ different distributions. Additionally, we define F_{α_0} to be the distribution corresponding to $\alpha_0 = 1.1$. Let $m = \ell \cdot n^3$. We define ℓ different instances of length m, where the i-th instance, Seq_i, is given by i blocks with n^3 samples from $F_{\alpha_1}, \dots, F_{\alpha_i}$, and then the remaining variables are samples of F_{α_0}. This is,

$$\mathrm{Seq}_i: \quad \underbrace{F_{\alpha_1}, \dots, F_{\alpha_1}}_{n^3 \text{ copies}}, \ \dots, \ \underbrace{F_{\alpha_i}, \dots, F_{\alpha_i}}_{n^3 \text{ copies}}, \ \underbrace{F_{\alpha_0}, \dots, F_{\alpha_0}}_{m - i\cdot n^3 \text{ copies}}.$$

Take an arbitrary online algorithm ALG that uses only the first k moments of the distributions. Because all distributions have the same first k moments, the ℓ instances are indistinguishable before the sequence starts, and up to block j, all instances Seq_i with $i \geq j$ are indistinguishable. Thus, we can define β_j as the probability that ALG stops in block j, on instances Seq_i with $i \geq j$. Since on the instance Seq_ℓ all the β_j are well defined and correspond to disjoint events, we have that $\sum_{j=1}^{\ell}\beta_j \leq 1$. Applying Lemma 4.2 on each block and on the copies of F_{α_0}, we get that the expected reward collected by the algorithm on instance Seq_i is at most

$$\mathbb{E}\left[\mathrm{ALG}(\mathrm{Seq}_i)\right] \leq \sum_{j=1}^{i}\left(n^{\alpha_j}\cdot\beta_j + \mathrm{O}(1/n)\right) + (\ell - i)\cdot n^{1.1}(1+\mathrm{o}(1))$$
$$= \mathrm{O}\left(\ell \cdot n^{1.1}\right) + \sum_{j=1}^{i} n \cdot 2^{s+j}\cdot\beta_j$$
$$= \left(\mathrm{O}\left(\ell/n^{0.15}\right) + \sum_{j=1}^{i} 2^{j-i}\cdot\beta_j\right)\cdot n\cdot 2^{s+i}$$
$$\leq \left(\mathrm{O}\left(\ell/n^{0.15}\right) + \sum_{j=1}^{i} 2^{j-i}\cdot\beta_j\right)\cdot(1+\mathrm{o}(1))$$
$$\cdot\,\mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right],$$

where we used that $n \cdot 2^s \geq n^{1.25}$, as well as Lemma 4.2 to get that $\mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right] \geq n^{\alpha_i}\cdot(1+\mathrm{o}(1))$. The adversary takes the instance that minimizes the competitive ratio, which is smaller than the average competitive ratio of the ℓ instances. We have that

$$\frac{1}{\ell}\sum_{i=1}^{\ell}\frac{\mathbb{E}\left[\mathrm{ALG}(\mathrm{Seq}_i)\right]}{\mathbb{E}\left[\mathrm{OPT}(\mathrm{Seq}_i)\right]} \leq \frac{1}{\ell}\sum_{i=1}^{\ell}\left(\mathrm{O}\left(\frac{\ell}{n^{0.15}}\right) + \sum_{j=1}^{i} 2^{j-i}\beta_j\right)(1+\mathrm{o}(1))$$
$$\leq \left(\mathrm{O}\left(\frac{\ell}{n^{0.15}}\right) + \frac{1}{\ell}\sum_{j=1}^{\ell}\beta_j\sum_{i=0}^{\infty}\frac{1}{2^i}\right)(1+\mathrm{o}(1))$$
$$\leq \left(\mathrm{O}\left(\frac{\ell}{n^{0.15}}\right) + \frac{2}{\ell}\right)(1+\mathrm{o}(1)).$$

Since $\ell = \Theta(\log n)$, the theorem follows. □

5 Full Moment Knowledge ($k = \infty$)

The result of the previous section raises a natural question: if knowledge of a very large but finite number of moments, even one growing with n, is insufficient to improve the competitive ratio, what happens when the decision maker has access to *all* moments ($k = \infty$)? This section demonstrates that even in this full-moment-information scenario, the same logarithmic barrier persists. Naturally, the distributions involved in the construction are not moment-determinate, as otherwise we would be able to recover the distributions from the moment sequence.

Theorem 5.1. *For every $n \geq 2$ there exists a family of instances of the ∞-moment prophet inequality problem with n random variables such that for every (deterministic or randomized) stopping rule* ALG*, we have*

$$\mathbb{E}[\mathrm{ALG}] \leq \mathrm{O}\left(\frac{1}{\log n}\right)\mathbb{E}[\mathrm{OPT}].$$

To prove the Theorem 5.1, our argument extends the construction from the finite case (Section 4) to the limit. We first show that for each α and n, the sequence of distributions $\{F_{\alpha,n}^{(k)}\}_{k\geq 3}$ from the previous section has a convergent subsequence, as $k \to \infty$. We then use these limit distributions to construct a family of indistinguishable instances that forces a poor competitive ratio for any algorithm.

Our construction begins with the family of auxiliary distributions $\{\tilde{F}_n^{(k)}\}_{k\geq 3}$ introduced in Section 4. This sequence converges in distribution: for any fixed support point $x \in \{0, 1, w_2, \dots\}$, the probability mass assigned to x by $\tilde{F}_n^{(k)}$ converges as $k \to \infty$ (see Equation (3)). Since the point-wise limit of the probability mass functions is itself a valid probability mass function, the sequence weakly converges to a well-defined limit distribution, which we denote by $\tilde{F}_n^{\infty}$. The moments of this limit distribution form our *universal moment sequence*. Our goal is to show that for any $\alpha \in (1, 2)$ and $n \in \mathbb{N}$, the corresponding sequence $\{F_{\alpha,n}^{(k)}\}_{k\geq 3}$ has a convergent subsequence, so we can define a limit distribution $F_{\alpha,n}^{\infty}$ whose moments all match this common sequence.

To establish this, we first prove that the family $\{F_{\alpha,n}^{(k)}\}_{k\geq 3}$ is *tight*, i.e., for any $\varepsilon > 0$, there exists a compact set $[0, M_\varepsilon]$ such that, for every distribution in the family, the probability mass outside this set is less than ε.

Lemma 5.1. *For each $\alpha \in (1, 2)$ and $n \geq 2$, the family $\{F_{\alpha,n}^{(k)}\}_{k\geq 3}$ is tight.*

Proof. The proof relies on the uniform bounds for the probabilities $p_\ell^{\alpha,n,k}$ derived in Lemma 4.1, which show that $p_\ell^{\alpha,n,k} = q_\ell(1 + \mathrm{o}(1))$ where the $\mathrm{o}(1)$ term is independent of k. Observe that for any $L \in \mathbb{N}$, the probability mass on the tail $\{w_\ell\}_{\ell>L}$ is bounded by:

$$\sum_{\ell=L+1}^{k} p_\ell^{\alpha,n,k} \leq \sum_{\ell=L+1}^{k} q_\ell(1 + \mathrm{o}(1)) \leq 2 \sum_{\ell=L+1}^{\infty} n^{-(\ell^2-1)}.$$

This bound on the tail sum is independent of k and converges to zero as $L \to \infty$. Thus, for any $\varepsilon > 0$, there exists a large enough $L(\varepsilon)$ such that the tail probability is less than ε, and we can set $M_\varepsilon = w_{L(\varepsilon)}$. □

Prokhorov's theorem [22] establishes that a tight sequence of distributions must always have a subsequence that converges in distribution. Therefore, $\{F_{\alpha,n}^{(k)}\}_{k\geq 3}$ has a subsequence that weakly converges to a limit probability measure. For each α and n, we define $F_{\alpha,n}^{\infty}$ as one of such limits. We remark that, even though Prokhorov's theorem does not guarantee uniqueness, any such limit of a subsequence works for our construction. We now prove that $F_{\alpha,n}^{\infty}$ has the same sequence of moments as $\tilde{F}_n^{\infty}$. To this end, we prove that the moments of the sequence of distributions converge to the moments of the limit distribution, for which we need uniform integrability.

Lemma 5.2. *For every $\alpha \in (1, 2)$, every $n \geq 2$, and every $m \in \mathbb{N}$, the sequence $\{X_k^m\}_{k\geq 3}$, where $X_k \sim F_{\alpha,n}^{(k)}$, is uniformly integrable. Similarly, if $X_k \sim \tilde{F}_n^{(k)}$, the sequence is also uniformly integrable.*

Proof. By the de la Vallée-Poussin criterion [21] of uniform integrability, it suffices to show that for some $r > 1$, we have $\sup_k \mathbb{E}_{\alpha,n}^{(k)}[|X^m|^r] < \infty$, where $\mathbb{E}_{\alpha,n}^{(k)}$ is the expectation respect to $F_{\alpha,n}^{(k)}$. Using the uniform probability bounds from Lemma 4.1, the contribution from the large support points is:

$$\begin{aligned}
\mathbb{E}_{\alpha,n}^{(k)}[|X^m|^r] &\leq 1 + (1 + \mathrm{o}(1))) \cdot n^{(rm-2)\alpha+1} \\
&\quad + (1 + \mathrm{o}(1)) \cdot \sum_{\ell=3}^{k} (w_\ell)^{mr} p_\ell^{\alpha,n,k} \\
&\leq 2 \sum_{\ell=2}^{\infty} n^{2mr(\ell-1)} \cdot n^{-(\ell^2-1)} \\
&= 2 \sum_{\ell=2}^{\infty} n^{-\ell^2+(2mr)\ell-2mr+1}.
\end{aligned}$$

This series converges since, for any fixed m and r, the term $-\ell^2$ dominates in the exponent. Additionally, note that the bound is independent of k, so the condition holds. Moreover, note that if $X \sim \tilde{F}_n^{(k)}$, the same bounds hold without the $(1 + \mathrm{o}(1))$ term, so the sequence is also uniformly integrable in this case. □

Combining these results, we can demonstrate that the limit distributions have the desired property: although they are distinct for different values of α, they share the same moment sequence.

Lemma 5.3. *For each parameter $\alpha \in (1, 2)$, the entire sequence of moments of $F_{\alpha,n}^{\infty}$ is identical to the sequence of moments of $\tilde{F}_n^{\infty}$ and is therefore independent of α.*

Proof. Let $\{F_{\alpha,n}^{(k_j)}\}_{j\geq 1}$ be a subsequence that converges weakly to $F_{\alpha,n}^{\infty}$, guaranteed to exist by Lemma 5.1. Since the sequence $\{X_{k_j}^m\}_{j\geq 1}$ is uniformly integrable for each m (Lemma 5.2), the moments of the sequence converge to the moments of the limit distribution, i.e., for all $m \in \mathbb{N}$,

$$\mathbb{E}_{\alpha,n}^{\infty}[X^m] = \lim_{j\to\infty} \mathbb{E}_{\alpha,n}^{(k_j)}[X^m],$$

where $\mathbb{E}_{\alpha,n}^{\infty}$ denotes the expectation when $X \sim F_{\alpha,n}^{\infty}$ and $\mathbb{E}_{\alpha,n}^{(k_j)}$ denotes the expectation when $X \sim F_{\alpha,n}^{(k_j)}$. From the construction in Section 4, for any $k_j \geq m$, we have the identity

$$\mathbb{E}_{\alpha,n}^{(k_j)}[X^m] = \tilde{\mathbb{E}}_n^{(k_j)}[X^m],$$

where $\tilde{\mathbb{E}}_n^{(k_j)}$ is the expectation respect to $\tilde{F}_n^{(k_j)}$. The right-hand side is the m-th moment of the base distribution $\tilde{F}_n^{(k_j)}$, which converges to the m-th moment of $\tilde{F}_n^{\infty}$ as $j \to \infty$. Combining these limits shows that $\mathbb{E}_{\alpha,n}^{\infty}[X^m]$ is equal to the m-th moment of $\tilde{F}_n^{\infty}$ for all m, i.e., is independent of α. □

Finally, we show that the key properties of the expected maximum hold in the limit.

Lemma 5.4. *For $n \geq 2$ and $r = n^3$, if we draw r i.i.d. samples $X_1, \dots, X_r$ from the limit distribution $F_{\alpha,n}^{\infty}$, then we have*

(1) $\mathbb{E}\left[\max_{i=1,\dots,r} X_i\right] = n^{\alpha} \cdot (1 + o(1))$, *and*
(2) $\mathbb{E}\left[\max_{i=1,\dots,r} X_i \cdot \mathbb{I}\left(X_i > n^{\alpha}\right)\right] \le O\left(\frac{1}{n}\right)$.

Proof. The bounds hold because the underlying probability estimates from Lemma 4.1 are uniform in k. Therefore, the bounds established in the proof of Lemma 4.2 hold for any $F^{(k)}_{\alpha,n}$ (with large enough k) and thus carry over to the weak limit $F^{\infty}_{\alpha,n}$. □

Now, we conclude the proof of Theorem 5.1.

Proof of Theorem 5.1. The proof is a direct adaptation of the one for Theorem 4.1. We construct $\ell = \Theta(\log n)$ different instances, where the i-th instance, Seq_i, consists of blocks of n^3 i.i.d. samples from the limit distributions $F^{\infty}_{\alpha_1,n}, \dots, F^{\infty}_{\alpha_i,n}$, followed by samples from a fixed distribution $F^{\infty}_{\alpha_0,n}$.

By Lemma 5.3, all these distributions have the identical moment sequence, making all ℓ instances completely indistinguishable to the algorithm. The analysis proceeds exactly as in the proof of Theorem 4.1. The expected reward for the algorithm in instance Seq_i is bounded using the probabilities β_j of stopping in block j, and the expected prophet value is controlled by Lemma 5.4. The adversary chooses the instance that minimizes the competitive ratio, which, when averaged over all ℓ instances, is shown to be no better than $O(1/\ell) = O(1/\log n)$. □

6 Improved Guarantees Under Additional Assumptions

One difficulty in estimating the expected maximum arises from the potential for tail events, which are rare outcomes of significant magnitude. The following two subsections explore this direction by analyzing distinct structural constraints. These constraints represent different forms of tail control: the first imposes a direct condition on the shape of the distribution's hazard rate (Section 6.1); the second operates by limiting the growth rate of its moments (Section 6.2). These cases collectively shed light on the extent to which these varied mechanisms of tail event control allow for a more efficient estimation of the expected maximum.

6.1 MHR Distributions

In this section, we study our problem under the additional constraint that all distributions belong to the class of absolutely continuous distributions with *monotone hazard rate* (MHR). A non-negative random variable with distribution F and density f, supported in (a, b), has a monotone hazard rate if $f(t)/(1 - F(t))$ is non-decreasing on (a, b). For this class, we obtain the following improved guarantee using a single moment, i.e., for $k = 1$.

Theorem 6.1. *For independent, non–negative random variables $X_1, \dots, X_n$ with MHR distributions, using only the first moment of each distribution, the modification of Algorithm 1 that samples j uniformly at random from $\{0, \dots, \lceil \log_2 \log n \rceil - 1\}$ satisfies*

$$\mathbb{E}[\mathrm{ALG}] = \Omega\left(\frac{1}{\log \log n}\right)\mathbb{E}[\mathrm{OPT}].$$

Intuitively speaking, the MHR assumption rules out very heavy tails. In practice, this implies that we can tighten the upper bound $\mathrm{OPT} \le n\mu^{\star}$ from Observation 3.1 to a logarithmic one, which we establish in the following lemma.

Lemma 6.1. *If $X_1, \dots, X_n$ are independent, non-negative MHR and $\mu^{\star} = \max_{i=1,\dots,n} \mathbb{E}[X_i]$, then $\mathbb{E}[\mathrm{OPT}] \le \mu^{\star} H_n \le \mu^{\star}(1 + \log n)$, where $H_n = \sum_{j=1}^{n} 1/j$.*

Proof. We use the fact that MHR distributions are increasing-convex dominated by exponential distributions; see, e.g., [25, Theorem 3.A.55], namely, we have $\mathbb{E}[g(X)] \le \mathbb{E}[g(Y)]$ when g is convex, $\mathbb{E}[X] \le \mu$, and $Y \sim \mathrm{Exp}(1/\mu)$. We denote this relation as $X \le_{\mathrm{ic}} Y$.

For each $i = 1, \dots n$ we pick an independent exponential $E_i \sim \mathrm{Exp}(1/\mu^{\star})$. Since X_i is MHR and $\mathbb{E}[X_i] \le \mu^{\star}$, we have $X_i \le_{\mathrm{ic}} E_i$. Notice that for every $a \in \mathbb{R}$, the function $g(x) = \max\{a, x\}$ is increasing and convex. Therefore, we can iteratively apply this property for X_i with E_i to conclude that $\mathbb{E}[\mathrm{OPT}] = \mathbb{E}\left[\max_{i=1,\dots,n} X_i\right] \le \mathbb{E}\left[\max_{i=1,\dots,n} E_i\right]$. Finally, we have

$$\begin{aligned}
\mathbb{E}\left[\max_{i=1,\dots,n} E_i\right] &= \int_0^{\infty} \left(1 - \left(1 - e^{-x/\mu^{\star}}\right)^n\right) dx \\
&= \mu^{\star} \int_0^1 (1 - (1-u)^n) \cdot \frac{1}{u}\, du \\
&= \mu^{\star} \int_0^1 \frac{1 - (1-u)^n}{1 - (1-u)}\, du = \mu^{\star} \int_0^1 \sum_{j=0}^{n-1} (1-u)^j\, du \\
&= \mu^{\star} \cdot H_n,
\end{aligned}$$

where $H_n = \sum_{j=1}^{n} 1/j \le \log n + 1$. □

Proof of Theorem 6.1. Combine the logarithmic upper bound $\mathrm{OPT} \le \mu^{\star}(1 + \log n)$ from Lemma 6.1 with the analysis of Algorithm 1 (exactly as in Theorem 3.1), replacing the range $[\mu^{\star},\ n\mu^{\star}]$ with $[\mu^{\star},\ \mu^{\star} \cdot (1 + \log n)]$, so that the probability of hitting the right bucket becomes $\Omega(1/\log \log n)$. □

We remark that the MHR family can be extended to discrete distributions by using the discrete density, i.e., $\Pr[X = t \mid X \ge t]$. Unfortunately, the improved guarantee cannot be extended to discrete MHR distributions; in particular, the distribution we construct in Section 4 is a discrete MHR distribution, and thus the $O(1/\log n)$-impossibility cannot be overcome. It is worth noting that MHR distributions are moment determinate, which means they are uniquely defined by their sequence of moments. This means that if all moments of the distributions are known (i.e., for $k = \infty$), we can recover the underlying distributions, so we can get a 1/2-approximation. But here, we demonstrate that for this important class of distributions, with just a single moment, a guarantee that is significantly better than the general $\Omega(1/\log n)$ can be obtained.

6.2 Bounded Coefficient of Variation

For a non-negative random variable Z let $\mathcal{V}_2(Z) := \mathbb{E}[Z^2]/(\mathbb{E}[Z])^2$ be the *coefficient of variation* of Z. For a given k, define $j^{\star} = \mathrm{argmax}_{i=1,\dots,n} \mathbb{E}[X_i^k]$. We assume that the distribution $F_{j^{\star}}$ of $X_{j^{\star}}$ has bounded coefficient of variation for some high-order moment, namely, there exist $c > 0$, $k \ge c \log n$ and fixed constant $D \ge 1$ such that[1]

$$\mathcal{V}_2(X^k_{j^{\star}}) = \frac{\mathbb{E}\left[X^{2k}_{j^{\star}}\right]}{(\mathbb{E}\left[X^k_{j^{\star}}\right])^2} \le D. \tag{7}$$

Our main result of this section is the following.

[1] By Jensen's inequality, D cannot be smaller than 1.

Theorem 6.2. *Fix $\theta \in [0,1]$ and let $T = \theta(\mathbb{E}[X_{j^\star}^k])^{1/k}$. For the k-moment prophet inequality, if (7) holds for $k \geq c \log n$ for some $c > 0$, then the single-threshold algorithm* ALG *that accepts the first random variable above T satisfies*

$$\mathbb{E}[\text{ALG}] \geq \frac{\theta(1-\theta^k)^2}{e^{1/c} D} \cdot \mathbb{E}[\text{OPT}].$$

When $c \geq 1$, by taking $\theta = 1/2$ the approximation guarantee becomes $\Omega(1/D)$.

To prove the theorem, we require the following upper bound on the prophet's expected value.

Lemma 6.2. *For every $k \geq 1$, we have*

$$\mathbb{E}\left[\max_{i=1,\dots,n} X_i\right] \leq n^{1/k} (\mathbb{E}[X_{j^\star}^k])^{1/k}.$$

Proof. Jensen's inequality yields

$$\mathbb{E}[\max_i X_i] = \mathbb{E}\left[(\max_i X_i^k)^{1/k}\right] \leq (\mathbb{E}\left[(\max_i X_i)^k\right])^{1/k}.$$

Furthermore, $(\max_i X_i)^k \leq \sum_{i=1}^n X_i^k$, and thus $(\mathbb{E}\left[\max_i X_i^k\right])^{1/k} \leq (\sum_{i=1}^n \mathbb{E}[X_i^k])^{1/k} \leq n^{1/k} (\mathbb{E}[X_{j^\star}^k])^{1/k}$. □

Notice that, for $k \geq c \log n$ for some c, we have $n^{1/k} \leq e^{1/c}$, and thus, the upper bound above is useful for distributions where the term $(\mathbb{E}[X_{j^\star}^k])^{1/k}$ is bounded for some k. Recall that the decision maker has access to the first k moments of every random variable. Consider now the single-threshold algorithm that sets a threshold $T = \theta(\mathbb{E}[X_{j^\star}^k])^{1/k}$ and accepts the first random variable above that threshold. Conditioned on accepting a value above T, Lemma 6.2 guarantees that such a value is a θ-approximation to $\mathbb{E}[\max_i X_i]$. Thus, it suffices to lower bound the probability that this algorithm selects a value above T. To do this, we use the Paley-Zygmund inequality, which we state here for completeness: If Z is a non-negative random variable with finite variance, then for $\theta \in [0,1]$ we have $\Pr[Z \geq \theta \mathbb{E}[Z]] \geq (1-\theta)^2 (\mathbb{E}[Z])^2 / \mathbb{E}[Z^2]$. We are now ready to prove the theorem.

Proof of Theorem 6.2. Let $k \geq c \log n$ for some $c > 0$. For $\theta \in [0,1]$ and for $Z = X_{j^\star}^k$, the Paley-Zygmund inequality guarantees that

$$\begin{aligned}\Pr\left[X_{j^\star} \geq \theta(\mathbb{E}[X_{j^\star}^k])^{1/k}\right] &= \Pr\left[X_{j^\star}^k \geq \theta^k \mathbb{E}[X_{j^\star}^k]\right] \\ &\geq (1-\theta^k)^2 \frac{(\mathbb{E}[X_{j^\star}^k])^2}{\mathbb{E}[X_{j^\star}^{2k}]} \geq \frac{(1-\theta^k)^2}{D},\end{aligned}$$

where the last inequality follows from (7). By Lemma 6.2, we have $\mathbb{E}[\text{OPT}] \leq T e^{1/c}/\theta$, and thus

$$\mathbb{E}[\text{ALG}] \geq \frac{(1-\theta^k)^2}{D} T \geq \frac{\theta(1-\theta^k)^2}{e^{1/c}\, D} \cdot \mathbb{E}[\text{OPT}]. \qquad \square$$

We remark that the distributional assumptions of Theorem 6.2 can be slightly weakened. Instead of (7), assume that there exist $c > 0$, $k \geq c \log n$, $d \geq 2$ and fixed constant $D \geq 1$ such that

$$\mathcal{V}_d(X_{j^\star}^k) = \left(\frac{\mathbb{E}[X_{j^\star}^{k\,d}]}{(\mathbb{E}[X_{j^\star}^k])^d}\right)^{\frac{1}{d-1}} \leq D. \tag{8}$$

Under this assumption, we show the following theorem.

Theorem 6.3. *Fix $0 \leq \theta \leq 1$ and let $T = \theta\ (\mathbb{E}[X_{j^\star}^k])^{1/k}$. For the k-moment prophet inequality, if (8) holds for $d \geq 2$ and $k \geq c \log n$ for some $c > 0$, then the single-threshold algorithm* ALG *that accepts the first random variable above T satisfies*

$$\mathbb{E}[\text{ALG}] \geq \frac{\theta(1-\theta^k)^{\frac{d}{d-1}}}{e^{1/c} D} \cdot \mathbb{E}[\text{OPT}].$$

If $c \geq 1$ and we take $\theta = 1/2$, the approximation guarantee becomes $\Omega(1/D)$.

To show the above theorem, we will need to make use of a higher-dimensional analogue of the Paley-Zygmund inequality. While its proof is standard, we include it here for completeness.

Lemma 6.3 (High-Dimensional Paley–Zygmund Inequality). *Let $d \geq 2$. If Z be a non-negative random variable with finite d-th moment, then for every $\theta \in [0,1]$,*

$$\Pr[Z \geq \theta\, \mathbb{E}[Z]] \geq \left((1-\theta)^d \frac{(\mathbb{E}[Z])^d}{\mathbb{E}[Z^d]}\right)^{\frac{1}{d-1}}.$$

Proof. We have

$$\begin{aligned}\mathbb{E}[Z] &= \mathbb{E}[Z\ \mathbb{I}(Z \geq \theta \mathbb{E}[Z])] + \mathbb{E}[Z\ \mathbb{I}(Z < \theta \mathbb{E}[Z])] \\ &\leq \mathbb{E}[Z\ \mathbb{I}(Z \geq \theta \mathbb{E}[Z])] + \theta\, \mathbb{E}[Z],\end{aligned}$$

which implies that

$$(1-\theta)\mathbb{E}[Z] \leq \mathbb{E}[Z\ \mathbb{I}(Z \geq \theta \mathbb{E}[Z])]. \tag{9}$$

By Hölder's inequality, we have

$$\mathbb{E}[Z\ \mathbb{I}(Z \geq \theta \mathbb{E}[Z])] \leq \left(\mathbb{E}[Z^d]\right)^{1/d} (\Pr[Z \geq \theta \mathbb{E}[Z]])^{\frac{d-1}{d}}. \tag{10}$$

Combining (9) and (10) yields

$$\begin{aligned}\Pr[Z \geq \theta \mathbb{E}[Z]] &\geq \left(\frac{\mathbb{E}[Z\ \mathbb{I}(Z \geq \theta \mathbb{E}[Z])]}{\left(\mathbb{E}\left[Z^d\right]\right)^{1/d}}\right)^{\frac{d}{d-1}} \\ &\geq \left((1-\theta)^d \frac{(\mathbb{E}[Z])^d}{\mathbb{E}\left[Z^d\right]}\right)^{\frac{1}{d-1}}. \qquad \square\end{aligned}$$

We are now ready to prove Theorem 6.3.

Proof of Theorem 6.3. Let $k \geq c \log n$ for some $c > 0$. For $\theta \in [0,1]$ and $Z = X_{j^\star}^k$, Lemma 6.3 guarantees that

$$\begin{aligned}\Pr\left[X_{j^\star} \geq \theta(\mathbb{E}[X_{j^\star}^k])^{1/k}\right] &= \Pr\left[X_{j^\star}^k \geq \theta^k \mathbb{E}[X_{j^\star}^k]\right] \\ &\geq \left((1-\theta^k)^d \frac{(\mathbb{E}[X_{j^\star}^k])^d}{\mathbb{E}[X_{j^\star}^{k\,d}]}\right)^{\frac{1}{d-1}} \\ &\geq \frac{(1-\theta^k)^{\frac{d}{d-1}}}{D},\end{aligned}$$

where the last inequality follows from (8). Since $\max_i X_i \geq X_{j^\star}$, $X_{j^\star} \geq T$ implies $\max_i X_i \geq T$ which in turn implies that ALG selects a value. By Lemma 6.2, we have $\mathbb{E}[\max_{i=1,\dots,n} X_i] \leq T e^{1/c}/\theta$, and thus

$$\mathbb{E}[\text{ALG}] \geq \frac{(1-\theta^k)^{\frac{d}{d-1}}}{D} T \geq \frac{\theta(1-\theta^k)^{\frac{d}{d-1}}}{e^{1/c}\, D} \cdot \mathbb{E}[\text{OPT}]. \qquad \square$$

7 Discussion

Motivated by the question of optimal stopping under partial information, we analyzed the *k-moment prophet inequality* problem. Our results reveal a striking phenomenon regarding the informativeness of moments in this setting. While one might intuitively expect more moments to provide better guarantees, our work demonstrates a strong negative result: the tight competitive ratio is $\Theta(1/\log n)$ when only the mean is known ($k = 1$), and surprisingly, access to any number of moments, or even the entire sequence of moments ($k = \infty$), provides no improvement on the guarantee. Of course, the latter is not true if we restrict ourselves to moment-determinate distributions, as we would recover the full distributions from the moment sequence. However, even within this class, the logarithmic barrier is inescapable for any finite k (even if k depends on n). This is in stark contrast to what happens in the other models of partial information that were previously studied, such as the sample-based model and the inaccurate-prior model, where more information yields better optimal performance, approaching the best possible guarantees as the amount of information increases.

This evidence motivates asking what conditions on the instance would increase the informativeness of the moment information. In the prophet inequality setting, the prophet's reward $\mathbb{E}[\max_i X_i]$ is sensitive to the tail behavior of the distributions. Our result suggests that the limited informativeness of moments in the general case stems from their inability to resolve ambiguity in the tails of the distributions, which can lead to vastly different expected maxima.

We demonstrated that moment information can become more helpful when combined with extra structural assumptions that control tail behavior. We explored a few such conditions: restricting distributions to the MHR class; imposing a bounded coefficient of variation for a sufficiently high-order moment, which limits the growth rate of moments; and directly bounding the support of the distributions. Each of these assumptions, in its own way, reduces the tail ambiguity that the adversary can exploit, helping to reduce the gap with the full-knowledge case.

Other natural and important families of distributions with controlled tail behavior are the sub-gaussian family (distributions such that $\Pr[|X| \geq t] \leq 2\exp(-t^2/C^2)$ for some $C > 0$), and the sub-exponential family (distributions such that $\Pr[|X| \geq t] \leq \exp(-Ct)$ for some $C > 0$). Interestingly, the distributions we construct for finite k have bounded support, so they are both sub-exponential and sub-gaussian for all k. While this means that the $O(\log n)$ barrier cannot be beaten in these classes, it might still be possible to beat it if we allow the number of moments to depend on the parameter C of the definition of these families.

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